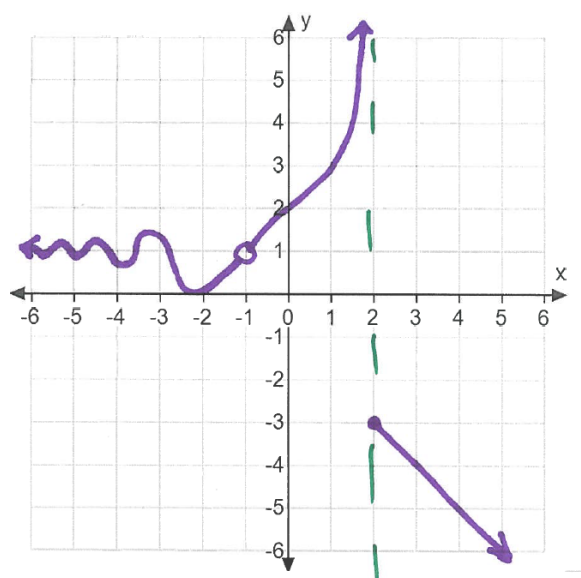


Problem 7

Problem 7

Problem 1 Use the graph of f to answer the questions below.



(a) State the domain of f .

Explanation. $(-\infty, -1) \cup (-1, \infty)$

(b) Find the following values or state "does not exist":

(i) $\lim_{x \rightarrow 2^-} f(x) =$

(ii) $\lim_{x \rightarrow 2^+} f(x) =$

(iii) $\lim_{x \rightarrow 2} f(x) =$

(iv) $\lim_{x \rightarrow -1} f(x) =$

(v) $\lim_{x \rightarrow 3} f(x) =$

(vi) $\lim_{x \rightarrow -\infty} f(x) =$

(vii) $f(-1) =$

Explanation. (i) $\lim_{x \rightarrow 2^-} f(x) = \infty$

Problem 7

- (ii) $\lim_{x \rightarrow 2^+} f(x) = -3$
- (iii) $\lim_{x \rightarrow 2} f(x)$ Does not exist
- (iv) $\lim_{x \rightarrow -1} f(x) = 1$
- (v) $\lim_{x \rightarrow 3} f(x) = -4$
- (vi) $\lim_{x \rightarrow -\infty} f(x) = 1$
- (vii) $f(-1)$ Does not exist

(c) *State the equation of any vertical asymptotes.*

Explanation. $x = 2$

(d) *State the equation of any horizontal asymptotes.*

Explanation. $y = 1$

(e) *Find the Intervals of Continuity.*

Explanation. $(-\infty, -1), (-1, 2), [2, \infty)$